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Time Signals and Power Spectral Density		

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1 Introduction and Objectives

The primary objective of this experiment is to practically examine and deepen the theoretical knowledge of digital amplitude modulation, specifically focusing on On-Off Keying (OOK). We aim to observe the transmission time signals, measure their corresponding power spectral densities (PSD), and investigate the impact of phase errors during coherent demodulation.

2 Preparation: Theoretical Signals

To establish a theoretical baseline prior to the practical measurements, the time signals and power density spectra for both BPSK and OOK were evaluated via MATLAB, assuming a symbol rate of $F_s = 4 \text{ kHz}$ and a carrier frequency of $f_T = 20 \text{ kHz}$.

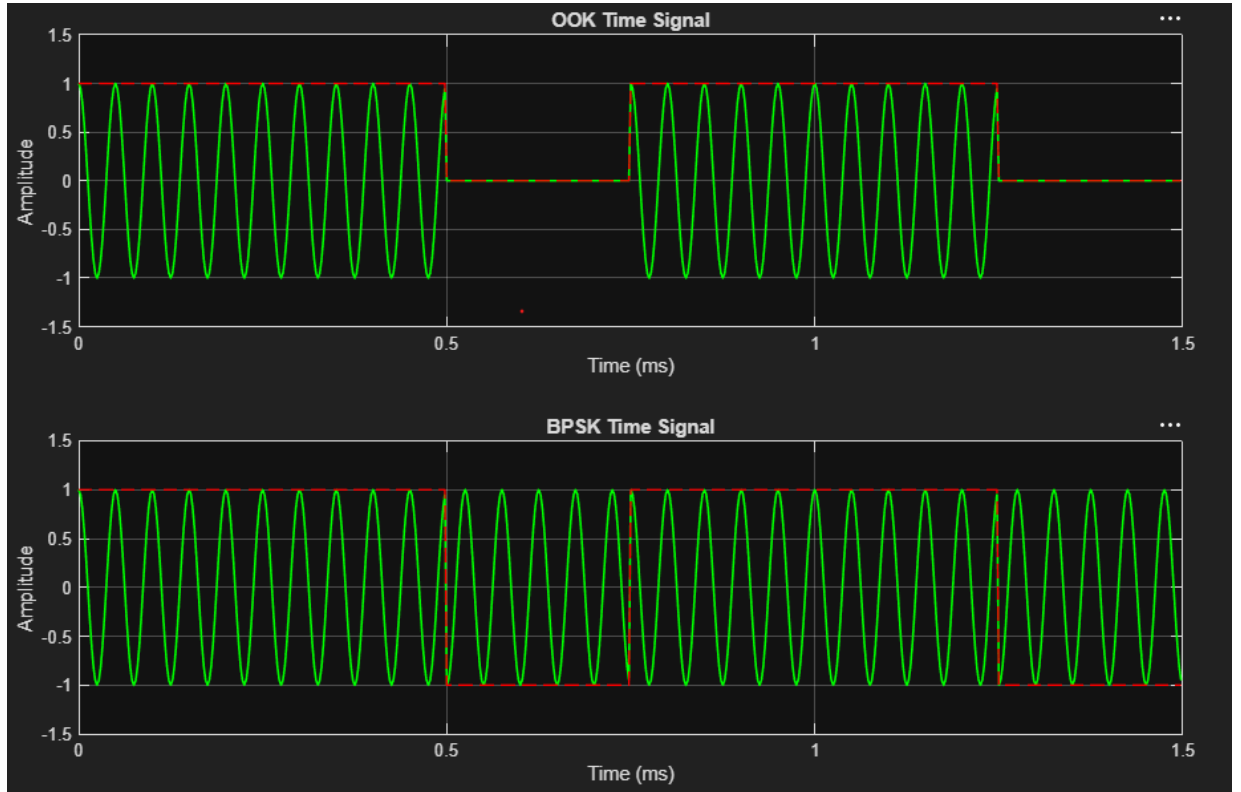


Figure 1: MATLAB plots of the theoretical time signals for BPSK and OOK ($F_s = 4 \text{ kHz}$, $f_T = 20 \text{ kHz}$).

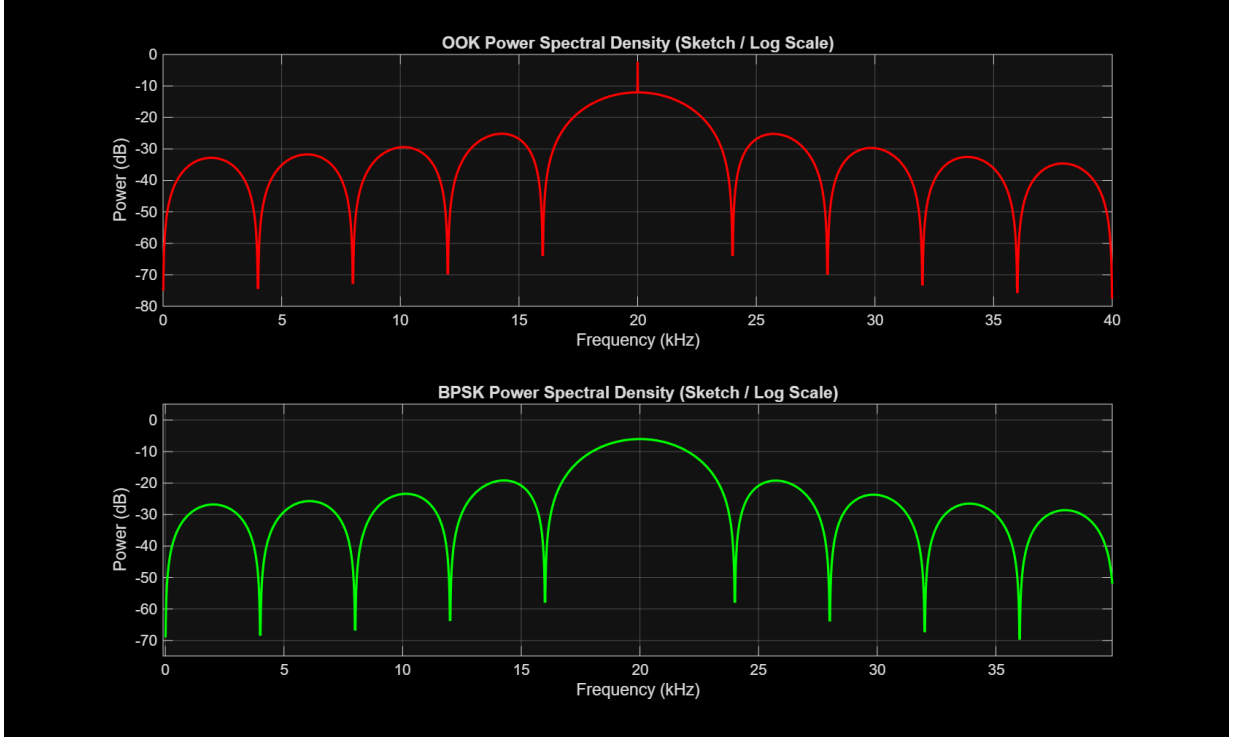


Figure 2: MATLAB plots of the theoretical power density spectra for BPSK and OOK.

2.1 Measurement Theory: UPV Analyzer vs. MATLAB

To properly compare theoretical simulations with physical measurements, it is necessary to understand how power spectral density (PSD) is calculated by different tools.

According to Parseval's theorem, the theoretical power density spectrum S_{uu} for a periodic signal, normalized to the frequency bin ΔF , is defined as:

$$S_{uu}[k \cdot \Delta F] = \frac{|\tilde{U}[k \cdot \Delta F]|^2}{R \cdot \Delta F} \quad (1)$$

MATLAB Implementation:

When utilizing MATLAB to compute the PSD, the standard `fft()` function computes an unnormalized Discrete Fourier Transform (DFT). To convert this raw output into a properly scaled PSD that matches the theoretical derivation, the squared magnitude must be explicitly normalized by the number of samples N and the sampling frequency F_A :

$$PSD_{\text{MATLAB}}[k] = \frac{|\text{fft}(u[n])|^2}{F_A \cdot N} \quad (2)$$

UPV Audio Analyzer Implementation:

In contrast, the UPV Audio Analyzer outputs a one-sided spectral distribution of the

effective (RMS) voltages, averaged over M samples. Because the UPV only measures input voltage and cannot directly determine the system resistance R , it displays the estimated power density scaled by a constant factor:

$$|U_{\text{eff},M}[k\Delta F]|^2 \approx \begin{cases} 2 \cdot S_{uu,M}[k\Delta F] \cdot R \cdot \Delta F & \text{for } k \neq 0 \\ S_{uu,M}[k\Delta F] \cdot R \cdot \Delta F & \text{for } k = 0 \end{cases} \quad (3)$$

Comparison:

While MATLAB requires explicit mathematical normalization ($F_A \cdot N$) to yield a true theoretical power density, the UPV inherently displays a scaled version of the power density in dBV. Because the system resistance R and the frequency resolution ΔF are constant during the measurement, the UPV accurately preserves the correct spectral shape. This allows for a direct visual comparison of envelope dynamics (such as lobe suppression and zero crossings) between our physical hardware measurements and theoretical MATLAB models, even if the absolute y-axis units differ.

3 Measurements and Analysis

3.1 Baseband Characterization: Pseudo-Random Number Generator (PZG)

Setup and Work Path: The initial phase involved characterizing the unmodulated baseband signal produced by the Pseudo-Random Number Generator (PZG). The clock frequency was set to 4 kHz. For time-domain measurements on the oscilloscope, the sequence length was set to $2^8 - 1$ to guarantee stable display synchronization. For frequency-domain measurements on the UPV Audio Analyzer, the sequence length was increased to $2^{16} - 1$. Two spectra were captured to observe the effects of signal averaging: a short-term spectrum ($N = 1$) and an averaged spectrum ($N = 10$).

Data Collected: As per the lab protocol, the specific UPV Audio Analyzer parameters used during the spectrum capture must be documented to contextualize the discrete periodogram:

- **Averaging Factor (N):** Measurements were taken at $N = 1$ (short-term) and $N = 10$ (averaged) to observe noise floor reduction.
- **Frequency Resolution (ΔF):** While the exact value for ΔF is displayed natively on the UPV interface, it is omitted from the raw PDF graph export. However, theoretically, the frequency resolution is defined as $\Delta F = F_A/L$ (where F_A is the sampling frequency and L is the FFT block length). Assuming a sampling frequency of $F_A = 40$ kHz, ΔF is strictly dependent on the UPV's internal block length configuration for that specific sampling rate.

In digital communications, a **symbol** represents a discrete signal state conveying data. Because OOK is binary, each symbol corresponds to exactly one bit of the pseudo-random sequence. The following figures display the transient behavior over 10 and 50 symbols, alongside their measured spectra.

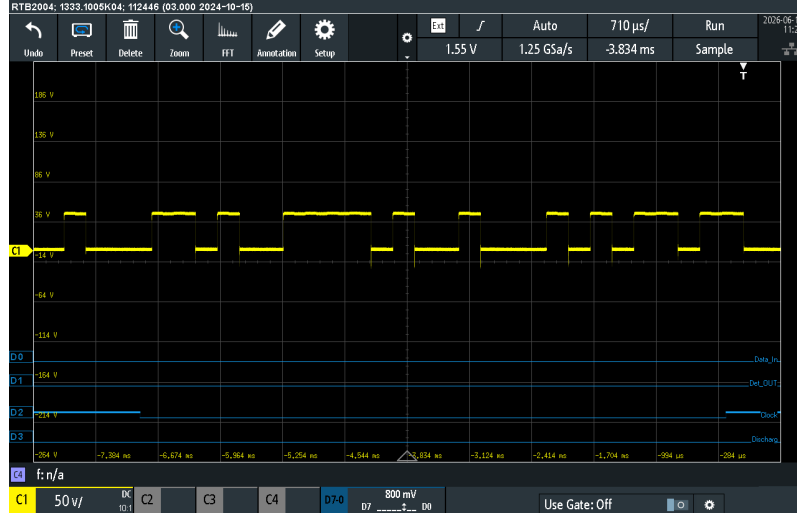


Figure 3: PZG time signal ($\tilde{10}$ symbols).

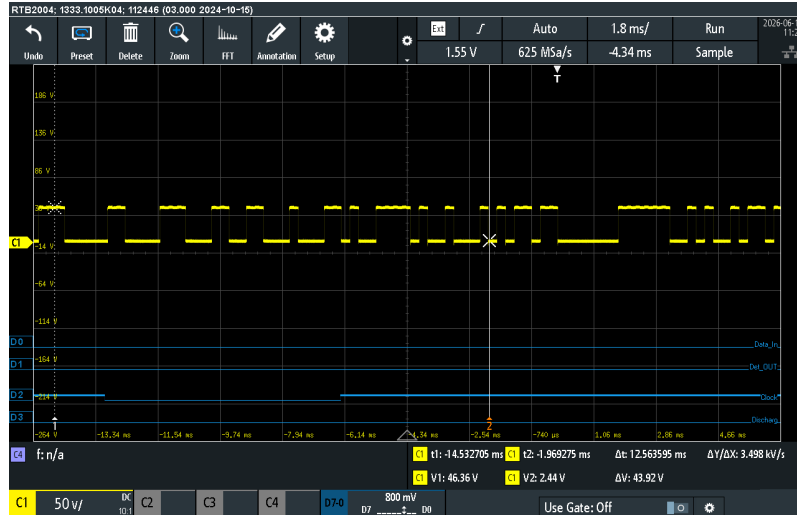
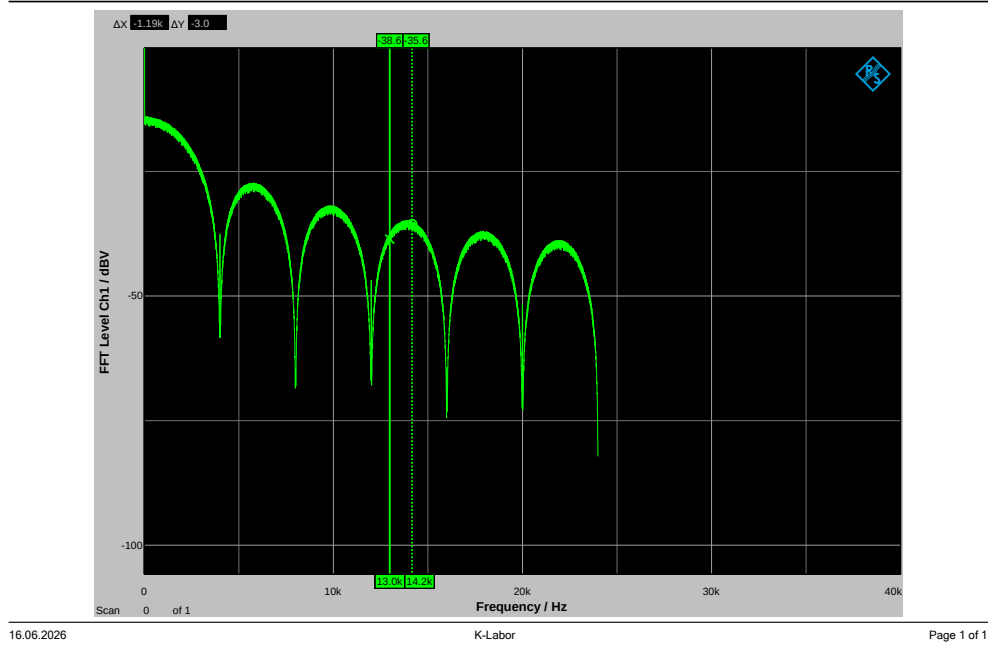
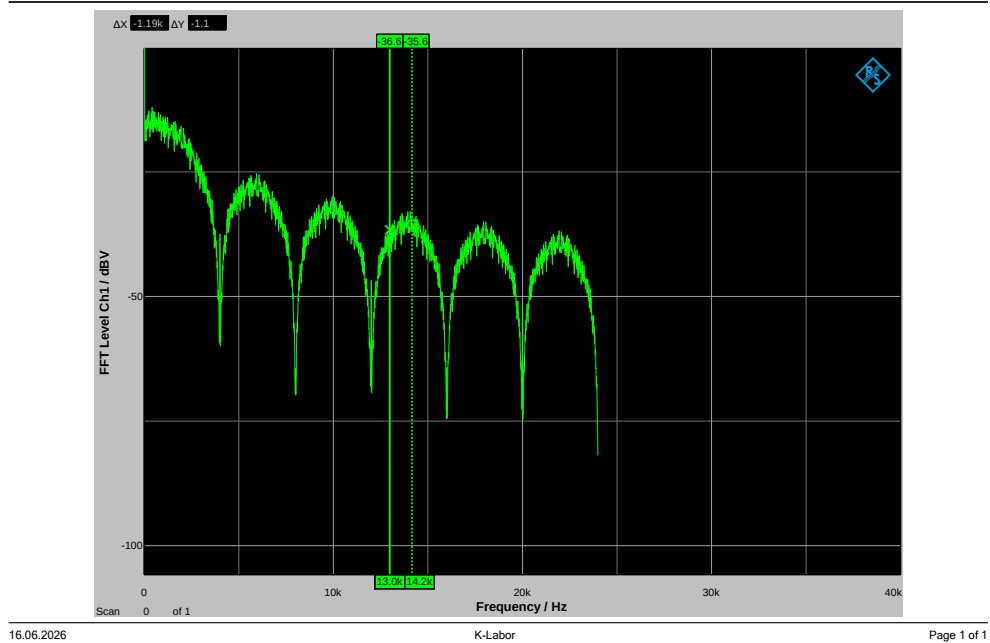


Figure 4: PZG time signal ($\tilde{50}$ symbols).

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Figure 5: Short-term power spectral density measurement ($N = 1$).

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Figure 6: Averaged power spectral density measurement ($N = 10$).

Analysis: Expected vs. Actual Results: The goal was to verify whether the PZG spectrum matches the theoretical sinc^2 shape expected from a rectangular pulse train at $F_s = 4 \text{ kHz}$.

- **Location of Zeros:** The power spectral density of a rectangular pulse is proportional to $\text{sinc}^2(fT_s) = \left(\frac{\sin(\pi fT_s)}{\pi fT_s}\right)^2$. The theoretical spectral zeros occur where the numerator equals zero (excluding the main lobe at $f = 0$):

$$\sin(\pi fT_s) = 0 \implies \pi fT_s = n\pi \implies f = \frac{n}{T_s} = n \cdot F_s \quad \text{for } n \in \mathbb{Z} \setminus \{0\} \quad (4)$$

Given our symbol rate of $F_s = 4 \text{ kHz}$, the expected zeros are at $f = n \cdot 4 \text{ kHz}$ (4 kHz, 8 kHz, 12 kHz, etc.). Our measurements distinctly confirm this exact location.

- **DC Component:** The measured DC peak is approximately -20 dB V . This prominent DC value confirms the unipolar nature of the physical TTL signal, which fluctuates between 0 V and a positive logic level, meaning its time-averaged amplitude cannot be zero.
- **Lobe Suppression and Sinc Envelope:** We measured a power drop of 5 dB from the main DC maximum to the first secondary lobe, and 12 dB to the second. To theoretically prove the ideal sinc^2 lobe suppression, we approximate the local maxima of the side lobes as occurring exactly halfway between the spectral zeros (at $f \approx 1.5F_s$ and $f \approx 2.5F_s$).

The power of the first side lobe relative to the main lobe (0 dB) is calculated as:

$$10 \log_{10} |\text{sinc}(1.5)|^2 = 10 \log_{10} \left(\frac{\sin(1.5\pi)}{1.5\pi} \right)^2 = 10 \log_{10} \left(\frac{1}{2.25\pi^2} \right) \approx -13.3 \text{ dB} \quad (5)$$

The power of the second side lobe is calculated as:

$$10 \log_{10} |\text{sinc}(2.5)|^2 = 10 \log_{10} \left(\frac{\sin(2.5\pi)}{2.5\pi} \right)^2 = 10 \log_{10} \left(\frac{1}{6.25\pi^2} \right) \approx -17.9 \text{ dB} \quad (6)$$

This proves that in an ideal, purely mathematical $\text{sinc}^2(x)$ spectrum, the first and second side lobes sit approximately 13.3 dB and 17.9 dB below the main lobe.

Conclusion: Zero locations confirm the sinc^2 shape. The sidelobe deviations are physically justified by measurement variance and non-ideal pulse shape, not hardware error.

3.2 Modulator System: On-Off Keying (OOK) and Carrier Synchronization

Setup and Work Path: The PZG signal was used to modulate a 20 kHz sinusoidal carrier ($0.3 \text{ V}_{\text{pp}}$) generated by Channel 1 of the Keysight Waveform Generator. The modulating

signal (Mod_IN) and the modulated output (RF_OUT) were observed simultaneously. To investigate phase coherence, the carrier frequency was intentionally shifted by 1 Hz, 10 Hz, and then fine-tuned in the parts-per-million (ppm) range.

Data Collected:

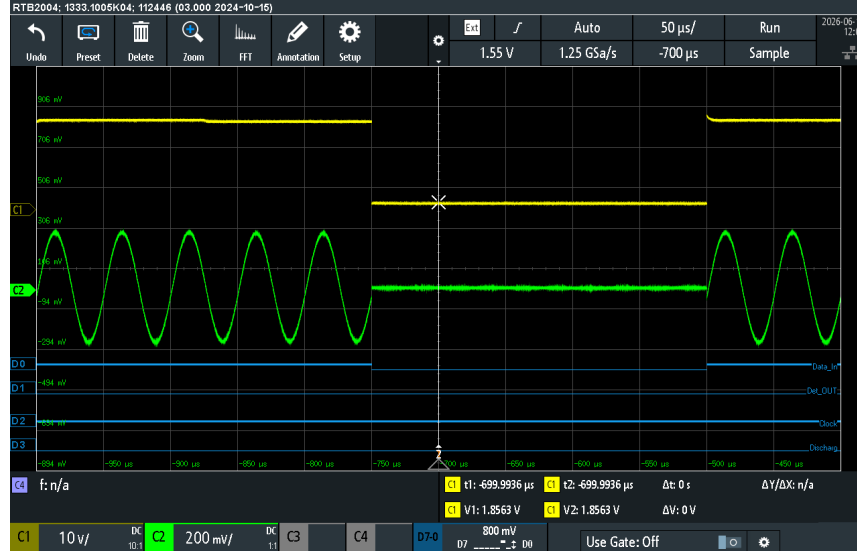


Figure 7: OOK time signal comparing Mod_IN and RF_OUT ($\tilde{2}$ symbols).

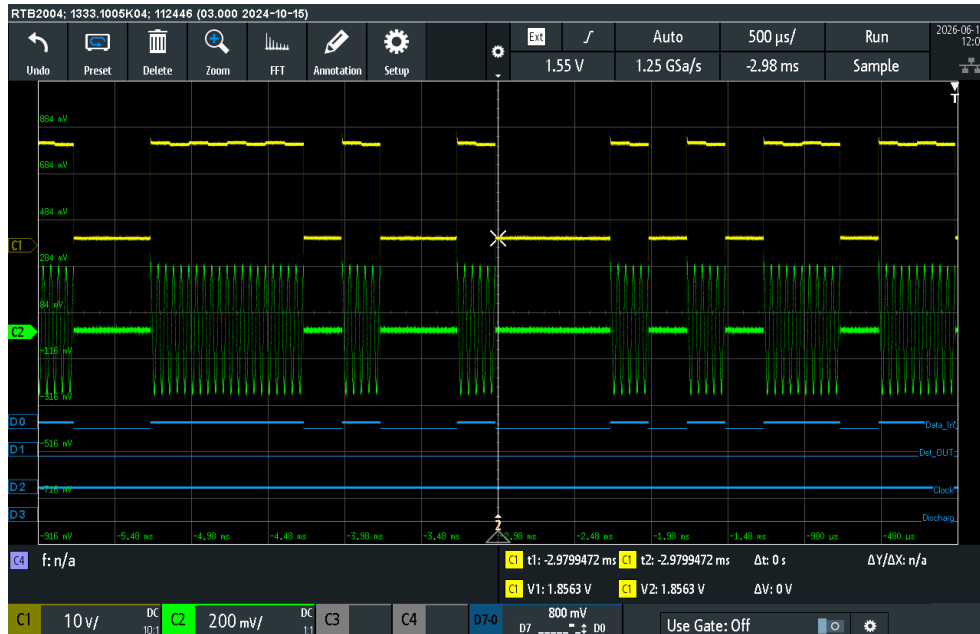


Figure 8: OOK time signal comparing Mod_IN and RF_OUT ($\tilde{24}$ symbols).

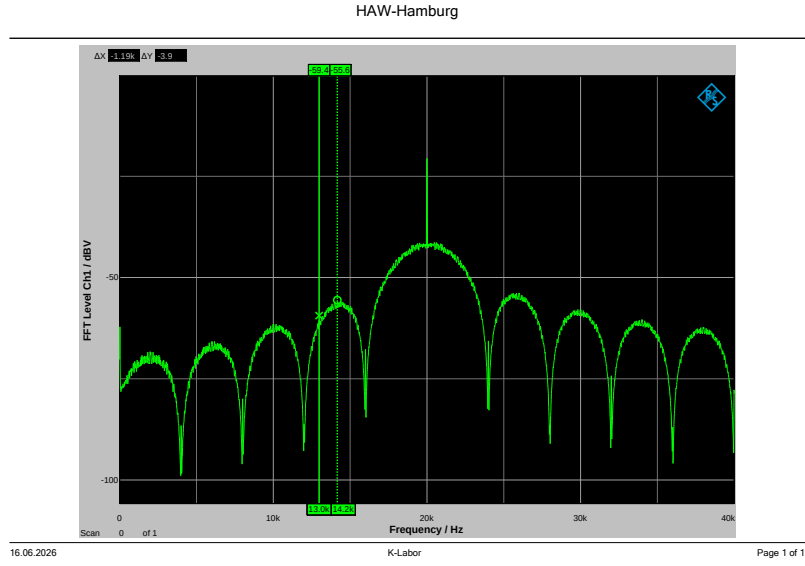


Figure 9: Power spectral density of the OOK modulated signal (0 Hz to 40 kHz).

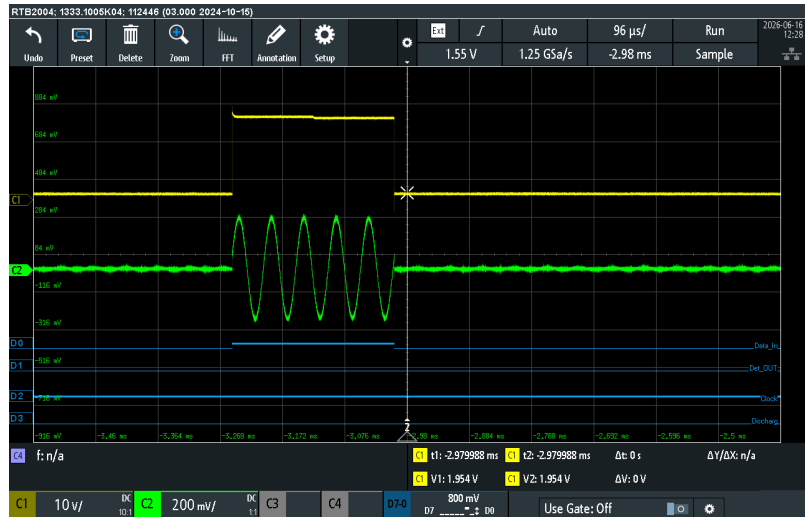


Figure 10: OOK signal with phase visually locked by fine-tuning the carrier to 19.999 820 kHz.

Analysis: The time-domain signals confirm expected OOK behavior: carrier present for logic high, suppressed for logic low. The spectrum shows the theoretical shifted sinc^2 sidebands around 20 kHz, with a prominent carrier spike – a direct consequence of the unipolar TTL baseband DC component being upconverted to f_T .

Frequency offsets of 1 Hz and 10 Hz caused visible phase drift on the oscilloscope. Stable lock was achieved at 19.999 820 kHz, reflecting minor oscillator tolerances between the PZG and function generator. Furthermore, as requested by the protocol, the carrier was

aggressively detuned to 19 kHz. As a result, the entire power spectral density (including the central carrier peak and sidebands) rigidly shifted down by exactly 1 kHz on the frequency axis. In the time domain, the individual carrier cycles within each rectangular symbol window became proportionally longer, visually confirming the direct dependence of the transmission signal on the f_T parameter.

Conclusion: Carrier spike, sideband structure, and frequency sensitivity all match theory. The lock frequency deviation is a hardware tolerance, not a system fault.

3.3 Channel Simulation: Distortion-Free Path Loss

Setup and Work Path: To model a physical transmission channel, the modulated signal (RF_OUT) was routed to the demodulator via an external ADD-VST-01 attenuator set to 20 dB. The internal 20 dB attenuation of the board was specifically avoided per the lab protocol to prevent undesirable noise coupling.

Data Collected:

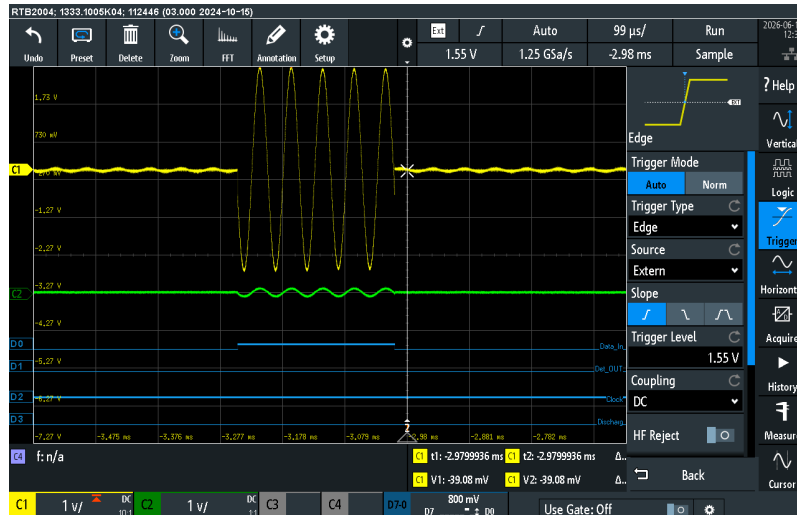


Figure 11: Oscilloscope capture demonstrating the 20 dB channel attenuation on the modulated signal.

Analysis: Expected vs. Actual Results: As desired, the oscilloscope capture demonstrates a pure attenuation of the transmitted carrier amplitude. The signal envelope shape and phase remain intact, successfully simulating a distortion-free path loss prior to entering the receiver.

3.4 Demodulator and Mixer Integration

3.4.1 Coherent Signal Downmixing (Zero Phase Error)

Setup and Work Path: At the receiver side, the attenuated signal was fed into the Aux input of the demodulator mixer. Channel 2 of the Keysight Waveform Generator provided a synchronized 20 kHz carrier to the Carrier-Demod_IN port. The phase difference was initially set to 0° (perfect coherent demodulation), and the downmixed signal was measured at Demod_OUT.

Data Collected:

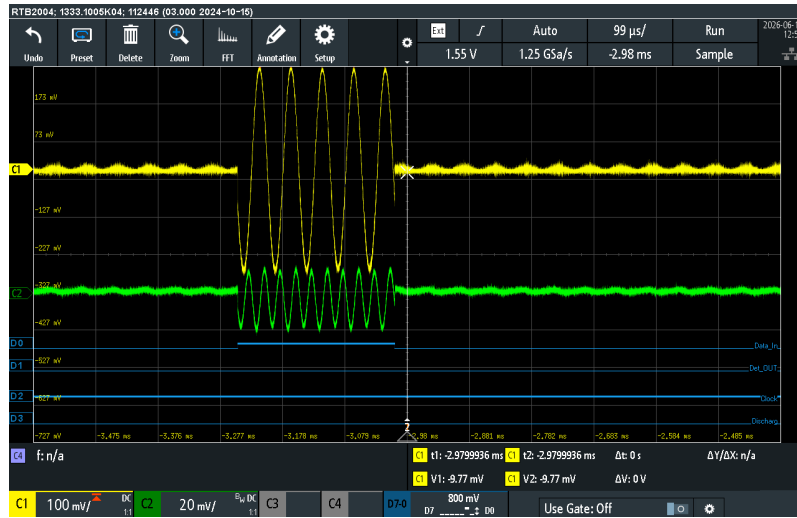


Figure 12: Time signals demonstrating the coherently downmixed signal at Demod_OUT.

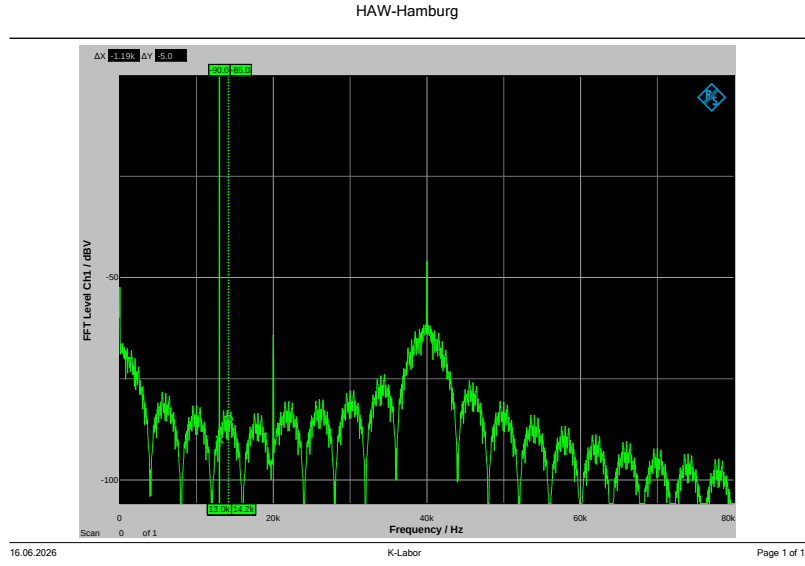


Figure 13: Power spectral density of the coherently demodulated signal (0 Hz to 80 kHz).

Analysis: Expected vs. Actual Results: The time-domain signal (Figure 12) and the spectrum (Figure 13) show successful downmixing. The spectrum reveals the desired shift of the signal energy back to the baseband (centered at 0 Hz), while the sum-frequency components emerge at $2\omega_T$ (40 kHz).

3.4.2 Baseband Signal Reconstruction via Low-Pass Filtering

Setup and Work Path: To recover the original baseband data and eliminate the high-frequency mixing artifacts, the Demod_OUT signal was passed through a custom K-Labor low-pass filter with a cutoff frequency of $f_g \approx 8$ kHz.

Data Collected:

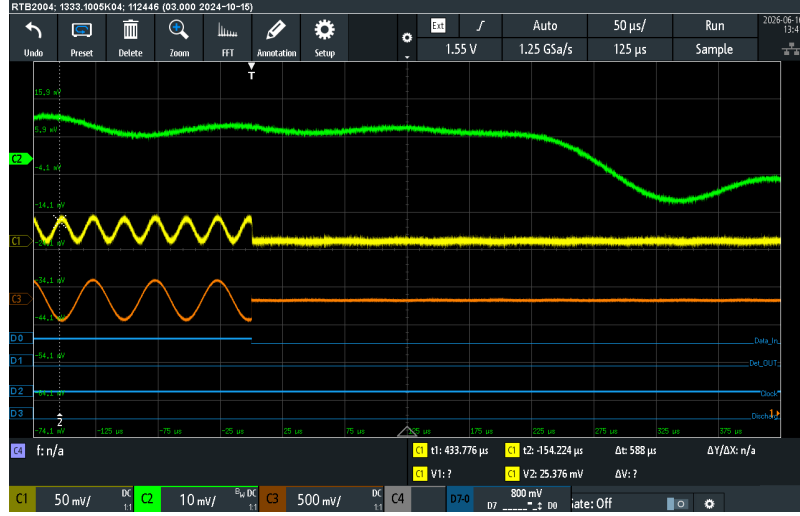


Figure 14: Signal reconstruction at the low-pass filter output ($\tilde{2}$ symbols).



Figure 15: Signal reconstruction at the low-pass filter output ($\tilde{24}$ symbols).

Analysis: Expected vs. Actual Results: As shown in Figures 14 and 15, the low-pass filter successfully rejects the $2\omega_T$ carrier components. The smoothed output cleanly traces the envelope of the original PZG data, achieving the desired goal of full signal reconstruction.

3.4.3 Impact of Phase Error on Coherent Demodulation

Setup and Work Path: In practical systems, phase synchronization is rarely perfect. To quantify the effect of a phase error ($\Delta\phi$), the receiver carrier phase was manually shifted from 0° to 180° in 10° increments. For the chosen OOK scheme, the DC voltage of

Demod_OUT was measured at each step using the UPV Audio Analyzer in RMS Voltmeter mode, and plotted via MATLAB. Spectra were also recorded at 45° and 90° .

Data Collected: Table ?? details the raw DC voltage measurements recorded for each 10° phase increment.

Table 1: Measured DC voltage at Demod_OUT as a function of phase error for OOK.

Phase Error $\Delta\phi$ [$^\circ$]	Measured DC Voltage [V]
0	7.35
10	7.15
20	6.75
30	6.15
40	5.30
50	4.35
60	3.25
70	2.05
80	0.80
90	-0.50
100	-1.80
110	-3.00
120	-4.25
130	-5.30
140	-6.15
150	-6.90
160	-7.40
170	-7.65
180	-7.70

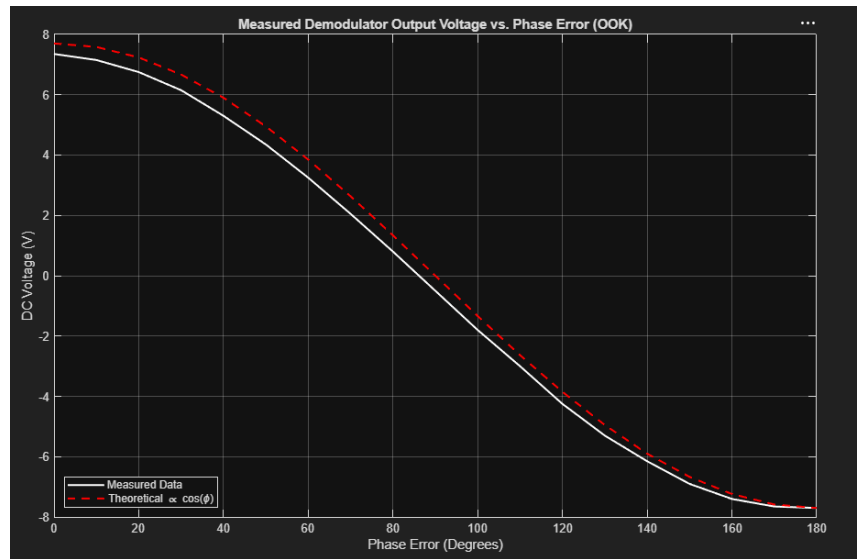


Figure 16: MATLAB plot of the measured demodulator output DC voltage vs. phase error.

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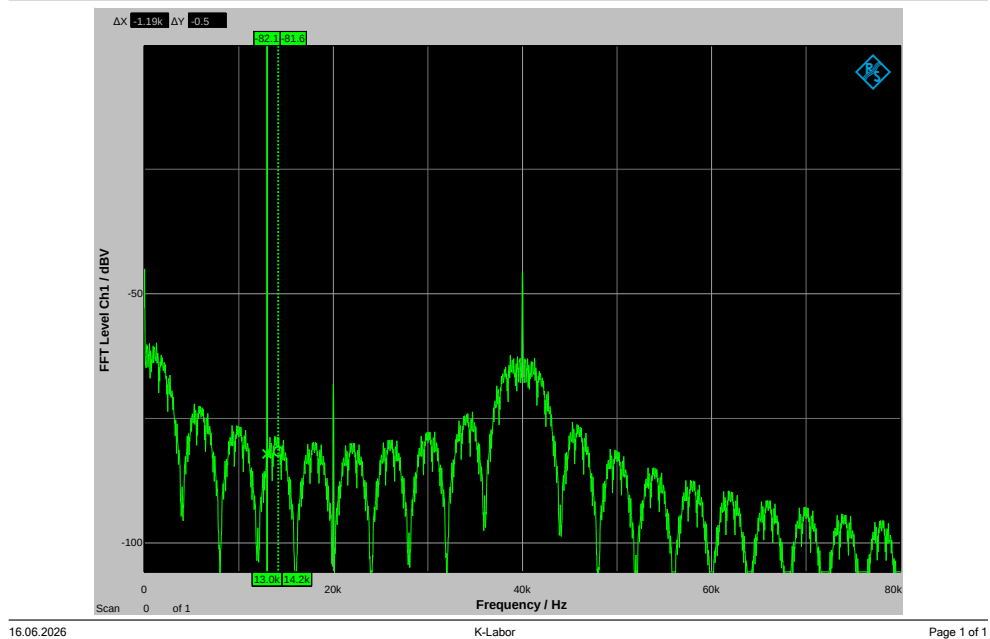


Figure 17: Power spectral density of Demod_OUT with a 45° phase error.

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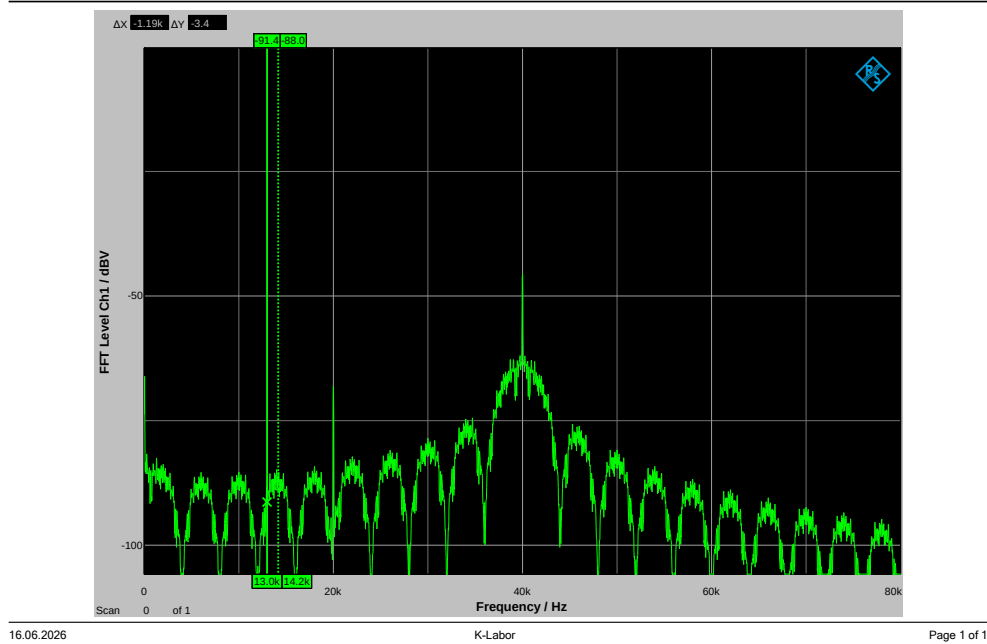


Figure 18: Power spectral density of Demod_OUT with a 90° phase error.

Analysis: Expected vs. Actual Results: The theoretical amplitude of a coherently demodulated signal degrades according to the cosine of the phase error ($\cos(\Delta\phi)$). The measured DC plot behaves exactly as desired, tightly tracking the theoretical cosine envelope and decaying toward zero as the phase approaches 90°.

In the frequency domain, the 45° spectrum shows partial attenuation, while the 90° spectrum (Figure 18) confirms that when the carrier is fully orthogonal, the demodulated signal theoretically nulls. The PSD appropriately collapses into the system noise floor, validating the severe consequences of a 90° phase offset.

Comprehension Analysis: OOK vs. BPSK Measurement Differences: The protocol requires different measurement techniques for OOK and BPSK due to the nature of their baseband logic. The OOK baseband is unipolar (0 and $+V$). When demodulated, it retains a distinct time-averaged DC offset directly proportional to the amplitude. Therefore, reading the DC voltage accurately captures the $\cos(\Delta\phi)$ attenuation. Conversely, BPSK utilizes a bipolar baseband sequence ($-V$ and $+V$). Because it is highly symmetric, its time-averaged DC voltage is always zero, regardless of phase error. To measure BPSK amplitude degradation, one must instead measure the AC RMS voltage of the recovered signal after the low-pass filter.

4 Evaluation and Conclusion

This experiment provided profound insights into the physical realities of digital amplitude modulation, bridging the gap between ideal mathematical models and practical hardware implementation. Rather than summarizing the chronological procedure, the following points represent the core conceptual takeaways, critical understandings, and "exam notes" derived from observing the discrepancies between theory and reality.

Core Understandings and Exam Notes

- The Physical Reality of Baseband Spectra:** While pure mathematical theory dictates that a rectangular pulse yields a perfect $\text{sinc}^2(f)$ power spectral density with an exact 13.3 dB drop to the first side lobe, physical TTL signals deviate from this.
Key Takeaway: Real-world signals possess finite rise and fall times, as well as minor ringing. These hardware imperfections alter the high-frequency spectral roll-off, meaning physical signals will rarely match the infinite, instantaneous transitions of a pure mathematical sinc function.
- Instrument Measurement Paradigms:** The UPV Audio Analyzer does not output normalized theoretical PSD directly; it measures input voltage and displays scaled effective (RMS) voltages in dBV.
Key Takeaway: An instrument can only display what it physically measures. When comparing lab data to MATLAB FFT models, one must account for windowing, averaging ($N = 10$), and the inherent scaling factors of the specific measurement instrument.
- The Carrier Peak (OOK vs. BPSK):** The OOK modulated spectrum features a massive, distinct peak exactly at the carrier frequency (20 kHz).
Key Takeaway: This occurs because OOK utilizes a unipolar baseband (0 and $+V$) containing a strong DC component. Multiplying a DC offset by a carrier sine wave yields a continuous pure sine wave component. In contrast, BPSK uses a symmetric, zero-mean bipolar baseband ($-V$ and $+V$), resulting in a fully suppressed carrier with no such central peak.
- Phase Coherence is Absolute:** Coherent demodulation is highly unforgiving. Even a 1 Hz deviation in the carrier frequency destroys phase locking and visual synchronization on the oscilloscope.
Key Takeaway: Hardware oscillators have physical tolerances. Achieving true mathematical coherence required fine-tuning the carrier to exactly 19.999 820 kHz to compensate for the parts-per-million deviation of the independent PZG clock.
- The Cosine Law of Phase Error:** If the local oscillator at the receiver is out of phase with the incoming transmission by $\Delta\phi$, the demodulated signal amplitude degrades proportionally to $\cos(\Delta\phi)$.

Key Takeaway: Phase error directly destroys data integrity. At exactly 90° , the signals are orthogonal, the cosine is zero, and the recovered data is completely nullified into the system noise floor.

- **Demodulation Measurement Methodology:** The fundamental nature of a modulation scheme dictates how its degradation must be measured.

Key Takeaway: Because OOK is unipolar, its amplitude degradation can be easily and accurately tracked via the time-averaged DC voltage. Because BPSK is bipolar (symmetric around 0 V), its time-averaged DC voltage is always zero regardless of signal strength. Therefore, phase error attenuation in BPSK must be measured using AC RMS voltage.

Ultimately, the lab demonstrated that digital communication is not just about the mathematics of modulation, but about managing hardware imperfections, synchronization tolerances, and understanding the physical implications of unipolar versus bipolar logic on the frequency spectrum.